



Figure 1: The Slicer welcome GUI

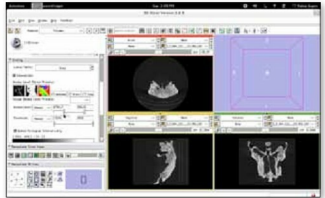


Figure 3: Changed orientation in 'Four-up Layout'

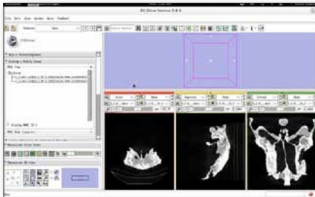


Figure 2: Scanned data of a Euoplocephalus skull (an ankylosaur dinosaur)



Figure 4: 'Green slice only' layout

3. Now click on the same menu and click on 'Green slice only layout', and the third image will be shown, filling up the whole window.

Let's modify the three images. The tool-bar at the top left contains a drop-down menu next to *Modules*. Select *Volumes*. You'll find various parameters to adjust the image rendering, but to put it simply, just use the slider in the display part of the volume accordion to obtain a sharp image, as shown in Figure 5.

You can move the slider above the CT image, and move back and forth through the skull. The number at the right (in mm) tells you how much of the image you have gone through.

Segmentation

You may even want to analyse some special part of the scanned image. The process by which you select useful parts of the data for further analysis is called segmentation. In Slicer, use a colour-coded label to indicate what is being segmented. For example, you may use blue for bones and green for tissue, as I've done here for this demonstration. Now go to the same drop-down menu next to *Modules* and select *Editor*, which will appear in the left panel. From the *Edit Selected Label Map* accordion choose from various labels. I selected *green* which represents tissue. Now we are ready for segmentation. There are various other ways to do this, but for beginners, let's use *Threshold*, as it is a quick method. On the left column, you

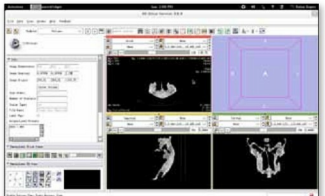


Figure 5: Obtaining a sharp image of the skull

will find a series of buttons. All have mouse hints; hover over them to find the button called *Threshold*. Click on it and you'll see a green pulsating image all over the CT scan monitor. Keep adjusting the slider bars until you get the skull part highlighted with green. Then hit *Apply*. The image is now segmented!

This is called *scene creation*. You can save it as a scene, and later load the scene directly for further analysis or to create a 3D model of the skull.

Now, from the fourth last button on the tool-bar, select *Conventional layout*. From the drop-down menu next to *Modules*, select *All Modules* and then *Model Maker* from this sub-menu. A new panel will pop up on the left. Set the